LALBA (α-Lactalbumin, UniProt P00709): Functional Annotation Report

Summary

Human **LALBA** (UniProt **P00709**; gene synonym *LYZL7*) encodes **α-lactalbumin (α-LA)**, a small (~14 kDa), secreted, calcium-binding whey protein that functions as the **regulatory "B" subunit of lactose synthase** (EC 2.4.1.22). α-Lactalbumin is not itself a catalytic enzyme. Instead, it is a **specificity-modifier protein**: it forms a 1:1 complex with the catalytic enzyme **β-1,4-galactosyltransferase-1 (β4Gal-T1)** and reprograms that enzyme's acceptor-substrate specificity. In the absence of α-LA, β4Gal-T1 preferentially transfers galactose from UDP-galactose onto *N*-acetylglucosamine (GlcNAc) to make *N*-acetyllactosamine. When α-LA binds, the complex instead transfers galactose onto free **glucose**, producing **lactose (Galβ1→4Glc)** — the dominant carbohydrate of milk. This is the protein's primary, physiologically essential function.

Structurally, α-lactalbumin is a paradox worth emphasizing: it is a **catalytically dead homolog of c-type lysozyme**. It shares 35–40% sequence identity and a nearly superimposable three-dimensional fold with hen egg-white lysozyme (a glycoside hydrolase, family GH22), yet it has lost the ability to bind and cleave glycosidic bonds. Evolution has repurposed the lysozyme scaffold into a glucose-presenting cofactor. α-LA is a **metalloprotein** with a strong, structurally critical Ca^{2+}-binding site (a distinctive non-EF-hand loop) and a separate, destabilizing Zn^{2+} site, giving it a well-known propensity to adopt a partially folded "molten-globule" state. It carries out its lactose-synthetic function in the **trans-Golgi lumen of lactating mammary secretory epithelial cells**, after which it is secreted into milk as the major whey protein, where it additionally serves as a source of essential amino acids (notably tryptophan and cysteine) and bioactive peptides for the nursing infant.

Two functional caveats frame the annotation. First, α-LA's lactose-synthetic role is **conditionally expressed**: it is switched on only in terminally differentiated, lactating mammary epithelium under the control of the lactogenic hormone triad (insulin + hydrocortisone + prolactin) acting through the prolactin→Stat5 pathway. Second, α-LA has a well-documented conformation-dependent **"moonlighting" activity**: in a partially unfolded state complexed with oleic acid it forms **HAMLET** (Human α-lactalbumin Made LEthal to

Tumor cells), a tumoricidal and antibacterial complex. This activity derives from the molten-globule conformation carrying a fatty acid — not from any native enzymatic function — and should be considered distinct from the protein's physiological role in milk sugar synthesis.

Identity Verification

The research target was fully verified. The gene symbol **LALBA** matches the UniProt description (Alpha-lactalbumin / Lactose synthase B protein / Lysozyme-like protein 7 = LYZL7). The organism is confirmed **Homo sapiens**. The protein-family and domain annotations (glycosyl hydrolase 22 family; Glyco_hydro_22, Lactalbumin, Lysozyme-like domains; Lys/PF00062) are **fully consistent** with the literature, which independently establishes the c-type-lysozyme fold and GH22 membership. There was **no ambiguity** and no conflation with a different gene — the published literature for human α-lactalbumin aligns precisely with the provided UniProt identity.

Key Findings

Finding 1 — α-Lactalbumin is the regulatory subunit of lactose synthase

UniProt P00709 (human LALBA; synonym LYZL7) is annotated as the "Lactose synthase B protein," and the primary literature confirms this directly. α-Lactalbumin forms a **1:1 heterodimeric complex** with the catalytic galactosyltransferase to constitute the functional **lactose synthase** enzyme system. Critically, recombinant human α-LA expressed in a heterologous host (transgenic tobacco) was shown to be **"fully active in the synthesis of lactose when combined with galactosyltransferase"** PMID: 9538226. This establishes that the α-LA polypeptide itself — independent of any mammalian post-translational modification — carries the information needed to confer lactose-synthetic capacity on the catalytic partner. The same work explicitly describes α-LA as **"a major milk protein… the regulatory subunit of lactose synthase."**

This is the central annotation: α-LA is a **regulatory/specificity subunit**, not a catalyst. It supplies substrate-specificity control rather than chemical catalysis.

Finding 2 — α-LA switches β4Gal-T1 acceptor specificity from GlcNAc to glucose

The molecular mechanism is now understood at atomic resolution. Crystal structures of lactose synthase at ~2 Å show that **α-LA promotes the binding of glucose to β4Gal-T1, thereby altering the enzyme's sugar-acceptor specificity from N-acetylglucosamine (GlcNAc) to glucose, which enables the complex to synthesize lactose** PMID: 11419947. α-LA holds the glucose acceptor in place via hydrogen bonding with the glucose O-1 hydroxyl in the acceptor site. Binding of α-LA is coupled to a **large conformational change** in β4Gal-T1 (a flexible loop closing over the active site).

Quantitatively, α-LA has a dramatic kinetic effect. It **increases β4Gal-T1's glucosyltransferase activity almost 30-fold** and lowers the Km for the glucose acceptor by a comparable factor PMID: 11485999. This converts a minor side-reaction of β4Gal-T1 (which has very low intrinsic activity toward glucose) into the dominant, biologically productive reaction. Later structural "snapshots" of β4Gal-T1 along its kinetic pathway, including a pentenary complex with α-lactalbumin, further detail how Mn^{2+} coordination and loop closure create the active ground-state geometry for catalysis PMID: 16497331.

Property	β4Gal-T1 alone	β4Gal-T1 + α-LA (lactose synthase)
Preferred acceptor	GlcNAc	Free glucose
Product	N-acetyllactosamine (Galβ1→4GlcNAc)	**Lactose (Galβ1→4Glc)**
Relative activity toward glucose	baseline (low)	**~30-fold increased**
Km for glucose acceptor	high	**~30-fold lowered**
Donor substrate	UDP-galactose	UDP-galactose
Metal cofactor	Mn^{2+}	Mn^{2+}

Finding 3 — α-LA is essential for lactose synthesis and milk production in vivo

The in vivo requirement for α-LA was established definitively by gene knockout. Mice homozygous for a targeted deletion of the α-lactalbumin gene are **"α-lac-deficient, produce reduced amounts of thickened milk containing little or no lactose, and cannot sustain their offspring,"** providing **"definitive evidence that α-lac is required for lactose synthesis and that lactose is important for milk production"** PMID: 7708733. Because lactose is osmotically active and is the principal osmole drawing water into the milk, loss of lactose synthesis produces a low-volume, abnormally viscous ("thickened") milk that cannot support the pups.

Crucially, the same study showed that **"females homozygous for the humanized allele lactate normally, indicating that human α-lac can replace murine α-lac."** This cross-species rescue demonstrates that the human P00709 protein performs the identical function and that the mechanism is evolutionarily conserved.

Finding 4 — α-LA has a c-type lysozyme fold and a unique Ca^{2+} site, but no hydrolase activity

High-resolution X-ray crystallography of baboon/human α-lactalbumin at 1.7 Å reveals a **two-domain architecture (an α-helical domain and a small β-sheet domain)** that is essentially the **c-type lysozyme fold** PMID: 2769757. A **tightly bound Ca^{2+} ion** sits in a distorted pentagonal-bipyramidal coordination geometry in a loop between the two domains. Importantly, this calcium-binding fold **"resembles an EF-hand only superficially and presumably has no evolutionary relationship with it"** — rather, it shows **"clear homology with the corresponding loop in c-type lysozyme (the 'parent' molecule of α-lactalbumin)"** PMID: 3785375. This is an example of a novel, convergent calcium site built on a lysozyme scaffold.

Despite the shared fold, α-LA has lost enzyme activity: **"Lysozymes bind and cleave the glycosidic bond linkage in sugars, whereas alpha-lactalbumin does not bind sugar but participates in the synthesis of lactose. Alpha-lactalbumin is a metallo-protein and binds calcium"** PMID: 10195284. This places α-LA firmly in the **glycosyl hydrolase 22 (GH22) family** by structure and ancestry (consistent with the Glyco_hydro_22 / Lysozyme-like domains in the UniProt annotation), while functionally it is a **catalytically dead, substrate-presenting cofactor**. The evolutionary transition from an enzyme (lysozyme) to a regulatory protein (α-LA) is a classic case study in protein engineering and evolution PMID: 3678590.

Finding 5 — A secondary Zn^{2+} site links metal binding to the molten-globule state

In addition to the strong structural Ca^{2+} site, α-lactalbumin possesses a **separate, strong Zn^{2+}-binding site** (association constant ~5×10^{5} M^{-1}). Occupancy of this and other secondary sites by zinc has a strongly destabilizing effect: **"The binding of zinc at secondary sites is accompanied by destabilization of the protein structure and progressive protein aggregation... reflected in a shift of the thermal denaturation transition temperature by more than 40 degrees"** PMID: 7945790. Fluorescence energy-transfer measurements place the strong Zn^{2+} (Co^{2+}) site **14–18 Å from the Ca^{2+} site, "in excellent agreement with recent crystallographic data for human alpha-lactalbumin."** Other heavy-metal ions (Pb^{2+}, Hg^{2+}) compete at the Ca^{2+} and Zn^{2+} sites and decrease thermal stability PMID: 8876980.

This metal-dependent conformational plasticity is mechanistically significant. Ca^{2+} binding **stabilizes** the native fold (and is required for efficient folding), whereas Zn^{2+} binding **destabilizes** it toward the partially folded molten-globule state. This "metal switch" underlies α-LA's famous conformational malleability, and it is directly relevant to the moonlighting HAMLET activity (Finding 6), which requires a partially unfolded conformation.

Finding 6 — Moonlighting: partially unfolded α-LA + oleic acid forms the tumoricidal HAMLET complex

α-Lactalbumin has a well-studied non-canonical activity. In a **partially unfolded (molten-globule) state complexed with oleic acid**, human α-LA forms **HAMLET** (Human α-lactalbumin Made LEthal to Tumor cells), which **"has potent tumoricidal activity"** and also kills some Gram-positive bacteria PMID: 34103518. A 39-residue α-helical peptide derived from α-LA forms oleate complexes (α1-oleate) that retain tumoricidal activity and reached safety/efficacy endpoints in a Phase I/II bladder cancer trial (NCT03560479). Related complexes include BAMLET (bovine α-LA–oleic acid) and liposome-delivered formulations.

Mechanistically, it is important to note that this cytotoxicity is **conformation- and lipid-dependent, not an intrinsic enzymatic property of the protein**: **"the protein portion of HAMLET, GAMLET, and the other HAMLET-like protein-oleic acid complexes is not the origin of their cytotoxicity"** PMID: 23580643. The oleic acid, delivered by the fluid protein scaffold, drives the toxicity. HAMLET is therefore best classified as a **moonlighting activity** distinct from LALBA's native physiological function in lactose synthesis.

Finding 7 — α-LA is the major human-milk whey protein and a source of amino acids and bioactive peptides

Beyond its catalytic-cofactor role, secreted α-LA is nutritionally central. An authoritative review notes that **"α-Lactalbumin is a whey protein that constitutes approximately 22% of the proteins in human milk and approximately 3.5% of those in bovine milk,"** and that **"within the mammary gland, α-lactalbumin plays a central role in milk production as part of the lactose synthase complex required for lactose formation, which drives milk volume"** PMID: 29617841. It is a rich source of tryptophan, lysine, branched-chain and sulfur amino acids. Upon digestion it **"is digested into smaller peptides with antimicrobial and prebiotic activities, immunostimulatory effect and acting as enhancers of mineral absorption"** PMID: 21335989. Its tryptophan richness modulates tryptophan/serotonin metabolism in developing infants, increasing circulating tryptophan and striatal serotonin in feeding models PMID: 40610456. These are secondary (nutritional/bioactive) roles of the secreted protein, downstream of its primary enzymatic-cofactor function.

Finding 8 — Expression is confined to lactating mammary tissue by lactogenic hormones (prolactin/Stat5)

α-Lactalbumin synthesis is a hallmark of terminal mammary differentiation. In mouse mammary explant cultures, **"functional differentiation was determined by the induction of casein and alpha-lactalbumin synthesis in the insulin, hydrocortisone, and prolactin (IFPrl)-treated explant cultures"** PMID: 3034921. α-LA is a pregnancy/lactation-specific gene product PMID: 6430550. Consistent with control by the prolactin→Stat5 axis, disrupting ErbB4→Stat5 signaling abolishes expression: in such mice **"whey acidic protein mRNA was reduced, and alpha-lactalbumin mRNA was undetectable"** at mid-lactation PMID: 10508857. This hormonal gating ensures that the lactose-synthase B-subunit — and therefore lactose production — is restricted to lactating mammary epithelium.

Finding 9 — Consolidated annotation

Convergent experimental, structural, genetic, and evolutionary evidence supports a single coherent annotation: **LALBA/α-lactalbumin is the regulatory B-subunit of lactose synthase (EC 2.4.1.22), acting in the mammary trans-Golgi**. (1) Isolated α-LA confers lactose-synthesis activity on galactosyltransferase [PMID: 9538226]; (2) crystal structures show it switches β4Gal-T1 acceptor specificity GlcNAc→glucose [PMID: 11419947, 11485999, 16497331]; (3) gene knockout abolishes milk lactose and lactation, rescued by the human gene [PMID: 7708733]; (4) it is a Ca^{2+}/Zn^{2+} metal-binding, catalytically inactive c-type lysozyme homolog (GH22) [PMID:

2769757, 3785375, 10195284, 7945790]; (5) expression is restricted to lactating mammary epithelium by prolactin/Stat5 lactogenic signaling [PMID: 3034921, 10508857]. A conformation-dependent HAMLET activity is distinct from the native function [PMID: 34103518, 23580643].

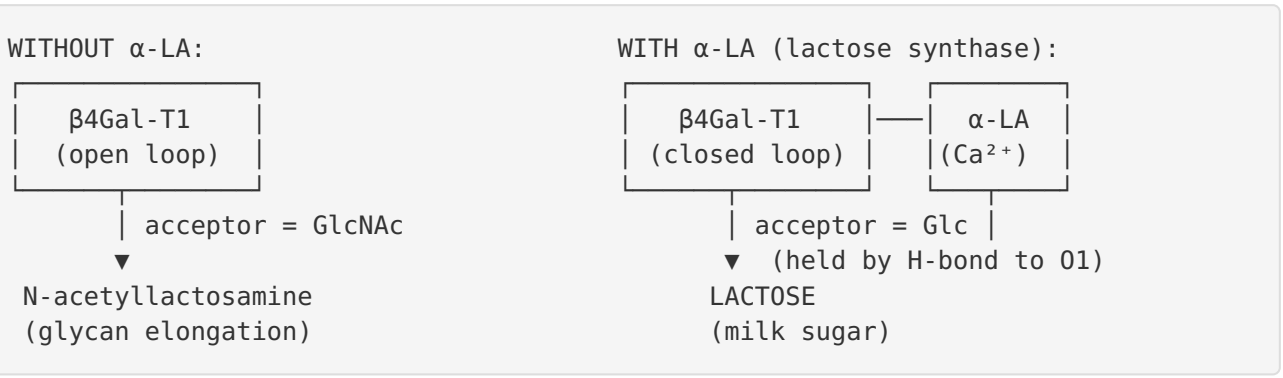
Mechanistic Model / Interpretation

The core reaction

Lactose synthase catalyzes, in the mammary Golgi:

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   UDP-galactose  +  D-glucose   ──(β4Gal-T1 + α-LA, Mn²⁺)──►   Lactose (Galβ1→4Glc)  +  UDP
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α-Lactalbumin does not participate in the chemistry. Its role is purely one of **substrate specificity control** and **acceptor presentation**:



The binding of α-LA (1) triggers closure of β4Gal-T1's flexible loop into the catalytically competent conformation and (2) creates a composite acceptor pocket that favors small, unmodified glucose over GlcNAc by hydrogen-bonding to the glucose O-1 hydroxyl. The net effect — a ~30-fold rate enhancement and ~30-fold drop in Km for glucose — rechannels the enzyme's output from glycan elongation to free-disaccharide synthesis.

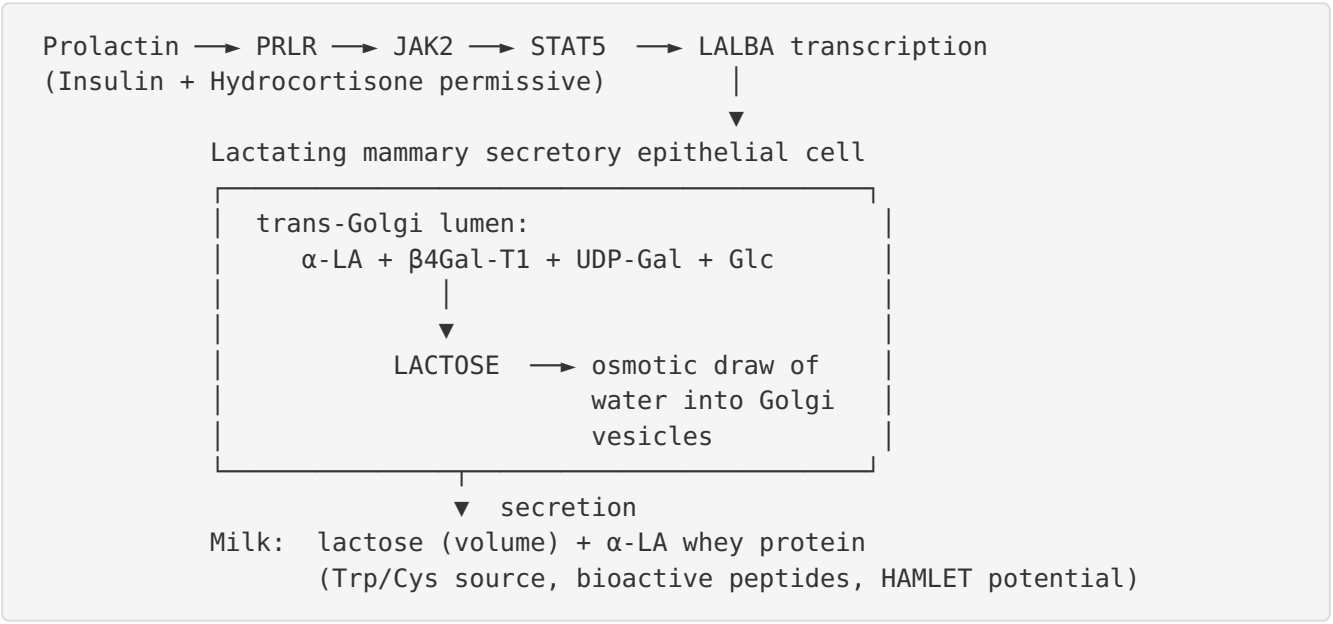
Evolution: enzyme → cofactor

α-Lactalbumin is a textbook example of evolutionary **neofunctionalization on a conserved scaffold**:

Feature	c-type lysozyme (ancestor)	α-lactalbumin (P00709)
Fold	α+β, two-domain lysozyme fold	Essentially identical fold
Sequence identity	—	35–40% to lysozyme
GH22 family	Yes (active glycosidase)	Yes (by homology; catalytically dead)
Substrate	Binds & cleaves peptidoglycan glycosidic bonds	Does **not** bind sugar; cannot hydrolyze
Metal	—	Strong structural Ca^{2+} (non-EF-hand loop) + destabilizing Zn^{2+} site
Function	Antibacterial hydrolase	Specificity modifier of β4Gal-T1

The lysozyme catalytic machinery was lost, and a high-affinity Ca^{2+} loop plus a glucose-presenting surface were gained, converting an enzyme into a regulatory partner protein.

Localization and physiological integration



α-LA carries out its catalytic-cofactor function **intracellularly, in the trans-Golgi lumen** of the secretory epithelial cell, co-localized with the membrane-anchored β4Gal-T1. The lactose it helps generate is osmotically active and draws water into secretory vesicles, which is why

lactose production **drives milk volume** — and why its loss yields low-volume, thickened milk in knockouts. The α-LA protein is then **secreted into milk**, where it takes on its secondary nutritional and bioactive roles extracellularly, in the infant gut.

Evidence Base

PMID	Title (abbrev.)	Contribution
9538226	Expression of human α-LA in transgenic tobacco	Recombinant α-LA is "fully active in lactose synthesis when combined with galactosyltransferase"; names it the regulatory subunit of lactose synthase (F1, F9)
11419947	Crystal structure of lactose synthase	Atomic mechanism: α-LA switches β4Gal-T1 acceptor specificity GlcNAc→glucose; large conformational change (F2, F9)
11485999	β4Gal-T1·LA crystal structure with UDP-Glc	Quantifies ~30-fold activity increase toward glucose acceptor (F2)
16497331	Structural snapshots of β4Gal-T1	Kinetic-pathway structures incl. α-LA pentenary complex; Mn^{2+}/loop dynamics (F2)
7708733	α-LA knockout / humanized rescue in mice	In vivo proof: no lactose, no viable lactation; human gene rescues (F3, F9)
2769757	Baboon α-LA structure at 1.7 Å vs lysozyme	Two-domain lysozyme fold; tightly bound Ca^{2+} (F4)
3785375	Novel Ca^{2+}-binding loop	Non-EF-hand Ca^{2+} site homologous to lysozyme loop (F4)
10195284	MD simulation of α-LA vs lysozyme	α-LA does not bind sugar; is a Ca^{2+} metalloprotein (F4)
3678590	From lysozyme to α-LA: engineering & evolution	Framing of enzyme→cofactor evolutionary transition (F4)
7945790	Co^{2+} binding to α-LA	Strong Zn^{2+} site; >40 °C destabilization; 14–18 Å from Ca^{2+} (F5)
8876980	Pb^{2+}/Hg^{2+} binding to α-LA	Heavy metals compete at Ca^{2+}/Zn^{2+} sites, reducing stability (F5)
34103518	Bladder cancer therapy with α1-oleate	HAMLET/α1-oleate tumoricidal; Phase I/II trial (F6)
23580643		

PMID	Title (abbrev.)	Contribution
	Mechanisms of HAMLET cytotoxicity	Cytotoxicity is conformation/lipid-dependent, not from protein per se (F6)
29617841	Applications for α-LA in human nutrition	Authoritative review: ~22% of human-milk protein; central lactose-synthase role drives milk volume (F7)
21335989	Biological effects of bovine milk fractions	Digestion yields antimicrobial/prebiotic peptides; enhances mineral absorption (F7)
40610456	α-LA, tryptophan & serotonin	Modulates Trp/serotonin metabolism in infants (F7)
3034921	Functional differentiation of mammary epithelium	Insulin+hydrocortisone+prolactin induce α-LA synthesis (F8)
10508857	ErbB4/Stat5 in mammary gland	Loss of ErbB4/Stat5 abolishes α-LA mRNA (F8)
6430550	Pregnancy-specific gene expression	α-LA is a pregnancy/lactation-specific differentiation product (F8)

Supporting cross-species / mechanistic context: The specificity-modifier concept is corroborated by the observation that α-LA can reprogram related glycosyltransferases from other organisms — e.g., it induces the snail *Lymnaea stagnalis* β4-GalNAcT to act on glucose >100-fold more efficiently, producing a lactose analog PMID: 8816760 — demonstrating that the α-LA interaction surface and the "acceptor-switching" mechanism are conserved features of this enzyme family. Early kinetic work on human milk galactosyltransferase also showed α-LA switches specificity only for monosaccharide acceptors PMID: 821753. Biophysical studies of α-LA folding (rapid molten-globule collapse, Ca^{2+}-accelerated folding) reinforce the metal-dependent conformational model underlying both native function and HAMLET formation PMID: 10329172, PMID: 16731974.

Limitations and Knowledge Gaps

- **Human-specific in vivo data are indirect.** The definitive genetic proof (knockout and lack of lactose) comes from mouse. The human protein's sufficiency is demonstrated by the

humanized-mouse rescue [PMID: 7708733] and by recombinant-protein activity assays [PMID: 9538226], but there is no human loss-of-function genetic model. Rare human LALBA variants and their phenotypic consequences are not well characterized in the reviewed literature.

- **LYZL7 synonym.** The UniProt synonym *LYZL7* (lysozyme-like 7) reflects the lysozyme homology, but the reviewed literature treats the protein uniformly as α-lactalbumin; any distinct "lysozyme-like 7" functional role outside lactation was not substantiated here.
- **Quantitative kinetics.** The ~30-fold activity/Km effects derive largely from in vitro bovine/human enzyme assays; precise human physiological rate constants in the intact Golgi are not directly measured.
- **HAMLET physiology.** Whether HAMLET-type complexes form in vivo during normal milk digestion remains unproven; HAMLET is primarily a therapeutic/biophysical phenomenon rather than a confirmed endogenous function.
- **Zn^{2+} regulation in vivo.** The destabilizing Zn^{2+} site is characterized biophysically; its physiological role (e.g., in modulating secretion, folding, or molten-globule formation in the secretory pathway) is not established.

Proposed Follow-up Experiments / Actions

1. **Human genetic association.** Survey human genomic/clinical databases (gnomAD, ClinVar) for LALBA loss-of-function or missense variants and correlate with lactation insufficiency or milk-composition phenotypes to provide direct human in vivo evidence.
2. **Cryo-EM/structural dynamics of the human complex.** Determine high-resolution structures of the intact human β4Gal-T1·α-LA·UDP-Gal·glucose complex (and trapped transition states) to refine the quantitative model of loop closure and glucose selection.
3. **Mutagenesis of the glucose-presenting interface.** Systematically mutate α-LA residues predicted to hydrogen-bond the glucose O-1 and measure effects on the GlcNAc→glucose switch and on lactose yield, mapping the minimal specificity-determinant surface.
4. **Secretory-pathway cell model.** Use mammary epithelial organoids with endogenous prolactin/Stat5 signaling to measure α-LA–dependent lactose flux and the osmotic/volume coupling in real time.
5. **Zn^{2+} and molten-globule physiology.** Test whether physiological Zn^{2+} fluxes in the secretory pathway modulate α-LA conformation, secretion efficiency, or endogenous fatty-acid-complex (HAMLET-like) formation.

6. **Metabolic/neurodevelopmental follow-up.** Extend the tryptophan→serotonin findings [PMID: 40610456] to controlled human infant-feeding studies to quantify the nutritional (as opposed to enzymatic-cofactor) contribution of secreted α-LA.

Report prepared from a 5-iteration autonomous investigation: 9 confirmed findings, 34 papers reviewed. Gene identity (LALBA / P00709 / Homo sapiens / GH22) verified against all literature with no ambiguity.
